

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1020

Pines 100 kV
May 18, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1020. The Interconnection Customer proposes a 143.9 MW wind facility interconnecting at the Pines 100 kV substation in the Ohio area. The request was received on July 06, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 440 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$28,219,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: MIGUEL LUJAN TAP NS LINE SUBSTATION / HOOKSINK 100kV
Contingency: BARAGA VLG / QUEEN CRK SUBSTATION 500kV
Loading: 341.4% (Rating: 128 MVA)
DFAX: 9.9%
Pre-project loading: 330.3%
2. Facility: MIGUEL LUJAN TAP NS LINE / WILLIAMS ST 100kV
Contingency: QUEENSBRIDGE / BURR OAK 100kV
Loading: 263.3% (Rating: 111 MVA)
DFAX: 8.2%
Pre-project loading: 252.6%
3. Facility: EVEREST / WEBSTER 100kV
Contingency: MIGUEL LUJAN TP NS LINE SUBSTATION / HOOKSINK 100kV
Loading: 252.8% (Rating: 98 MVA)
DFAX: 7.6%
Pre-project loading: 241.7%
4. Facility: LAWOTHER DR / CARO 100kV
Contingency: BARAGA VLG 500 kV/230 kV Autotransformer
Loading: 242.7% (Rating: 100 MVA)
DFAX: 7.5%
Pre-project loading: 232.0%
5. Facility: SCOTT CO / HOOKSINK 100kV line
Contingency: SCOTT CO SUBSTATION / BURR OAK 100kV
Loading: 226.3% (Rating: 209 MVA)
DFAX: 11.0%
Pre-project loading: 218.7%

6. Facility: EVEREST / QUEENSBRIDGE 100kV
Contingency: MIGUEL LUJAN TAP NS LINE SUBSTATION / HOOKSINK 100kV
Loading: 224.2% (Rating: 119 MVA)
DFAX: 7.6%
Pre-project loading: 215.0%
7. Facility: BUR OAK / QUEENSBRIDGE 100kV
Contingency: MIGUEL LUJAN TAP NS LINE / HOOKSINK 100kV
Loading: 223.1% (Rating: 133 MVA)
DFAX: 7.6%
Pre-project loading: 214.9%
8. Facility: CARO / MIAMI 100kV line
Contingency: BARAGA VLG 500/230kV Transformer Bank
Loading: 218.2% (Rating: 124 MVA)
DFAX: 7.5%
Pre-project loading: 209.5%
9. Facility: SCOTT CO / BURR OAK 100kV line
Contingency: SCOTT CO / HOOKSINK 100kV line
Loading: 216.3% (Rating: 182 MVA)
DFAX: 8.9%
Pre-project loading: 209.3%
10. Facility: WEBSTER / GENOA 100kV
Contingency: MIGUEL LUJAN TAP NS LINE / HOOKSINK 100kV
Loading: 207.4% (Rating: 114 MVA)
DFAX: 7.6%
Pre-project loading: 197.8%
11. Facility: WILLIAMS ST SUBSTATION / TAIT ELECTRIC GENERATING STA 100kV
Contingency: BARAA VLG 500/230 kV Transformer
Loading: 176.5% (Rating: 119 MVA)
DFAX: 5.8%
Pre-project loading: 169.4%
12. Facility: WILLIAMS T / GENOA 100kV
Contingency: MIGUEL LUJAN TAP NS LINE / HOOKSINK 100kV Line
Loading: 138.3% (Rating: 136 MVA)
DFAX: 7.6%
Pre-project loading: 130.3%
13. Facility: TAIT ELECTRIC GENERATING STA / HATCH 100kV
Contingency: BARAGA VLG 500-230 kV Auto
Loading: 138.0% (Rating: 132 MVA)
DFAX: 5.8%
Pre-project loading: 131.7%
14. Facility: FITCHBURG / WEST RIVERSIDE 100kV
Contingency: MIAMI / FIDDYMENT 100kV
Loading: 132.1% (Rating: 130 MVA)
DFAX: 9.3%
Pre-project loading: 121.7%
15. Facility: NORWICH 230/1kV XFMR
Contingency: BARRETT / NORWICH 500kV L
Loading: 119.2% (Rating: 995 MVA)
DFAX: 11.2%
Pre-project loading: 117.6%
16. Facility: EAST METHUEN / SWAN VLY 500kV
Contingency: MUSKOGEE / MONTPELIER ELECTRIC GENERATING STA 500kV

- Loading: 117.2% (Rating: 2,718 MVA)
DFAX: 17.8%
Pre-project loading: 116.2%
17. Facility: PATRICK / ROES CORNER 100kV Line
Contingency: BARAGA VLG 500-230 kV Autotransformer
Loading: 116.3% (Rating: 107 MVA)
DFAX: 17.1%
Pre-project loading: 93.2%
18. Facility: MUSKOGEE SUBSTATION / MONTPELIER ELECTRIC GENERATING STA SUBSTATION 500kV
Contingency: E METHUEN SUBSTATION / SWAN VALLEY SUBSTATION 500kV
Loading: 115.8% (Rating: 3,404 MVA)
DFAX: 9.4%
Pre-project loading: 115.4%
19. Facility: OMNI-HACKBERRY / NORWICH 500kV
Contingency: TREGO / BEAVER 500kV
Loading: 115.6% (Rating: 3,369 MVA)
DFAX: 31.5%
Pre-project loading: 114.2%
20. Facility: SWAN VALLEY / TREGO 500kV
Contingency: TROY ENERGY LLC / RICHLAND 500kV L
Loading: 114.1% (Rating: 4,453 MVA)
DFAX: 14.7%
Pre-project loading: 113.6%
21. Facility: MUSKOGEE / SWAN VALLEY 500kV
Contingency: EAST METHUEN / SWA VALLEY 500kV Line
Loading: 112.2% (Rating: 3,396 MVA)
DFAX: 8.4%
Pre-project loading: 111.8%
22. Facility: SAN VLY 500/230kV TX
Contingency: SWANVALLEY / TREGO 500kV
Loading: 112.0% (Rating: 1,622 MVA)
DFAX: 6.7%
Pre-project loading: 111.4%
23. Facility: IMPERIAL / CHAMBERLIN 230kV L
Contingency: HYLEBOS / RAVENNA 500kV
Loading: 111.8% (Rating: 1,299 MVA)
DFAX: 5.4%
Pre-project loading: 111.2%
24. Facility: STEVENS RD / CONESVILLE STA 230kV Line
Contingency: CONESVILLE STA / MUSKINGUM RIV 500kV
Loading: 111.7% (Rating: 791 MVA)
DFAX: 5.4%
Pre-project loading: 110.7%
25. Facility: 15TH AVENUE S / LUCENT 230kV
Contingency: OMNI-HACKBERRY / NORWICH 500kV
Loading: 111.5% (Rating: 616 MVA)
DFAX: 8.4%
Pre-project loading: 109.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MIGUEL LUJAN TAP NS LINE SUBSTATION / HOOKSINK 100kV	Rebuild line with higher-capacity conductor	\$2,055,000	18% (\$375,000)
MIGUEL LUJAN TAP NS LINE SUBSTATION / WILLIAMS ST 100kV	Rebuild line with higher-capacity conductor	\$3,028,000	24% (\$710,000)
EVEREST / WEBSTER 100kV	Rebuild line with higher-capacity conductor	\$1,708,000	23% (\$399,000)
LAWTHER DR / CARO 100kV	Rebuild line with higher-capacity conductor	\$2,199,000	42% (\$924,000)
SCOTT CO / HOOKSINK 100kV line	Rebuild line with higher-capacity conductor	\$2,408,000	23% (\$562,000)
EVEREST / QUEENSBRIDGE 100kV	Rebuild line with higher-capacity conductor	\$1,371,000	23% (\$320,000)
BURR OAK / QUEENSBRIDGE 100kV Line	Rebuild line with higher-capacity conductor	\$1,698,000	23% (\$397,000)
CARO / MIAMI 100kV line	Rebuild line with higher-capacity conductor	\$5,125,000	42% (\$2,153,000)
SCOTT CO / BURR OAK 100kV line	Rebuild line with higher-capacity conductor	\$5,702,000	23% (\$1,332,000)
WEBSTER SUBSTATION / GENOA 100kV	Rebuild line with higher-capacity conductor	\$1,707,000	23% (\$399,000)
WILLIAMS ST / TAIT ELECTRIC GENERATING STA 100kV	Rebuild line with higher-capacity conductor	\$2,835,000	24% (\$668,000)
WILLIAMS T / GENOA 100kV	Rebuild line with higher-capacity conductor	\$2,192,000	23% (\$512,000)
TAIT ELECTRIC GENERATING STA / HATCH 100kV	Rebuild line with higher-capacity conductor	\$1,525,000	24% (\$360,000)
FITCHBURG / WEST RIVERSIDE 100kV	Rebuild line with higher-capacity conductor	\$4,537,000	100% (\$4,537,000)
NORWICH 230/1kV XFMR	Replace transformer with higher-rated unit	\$12,480,000	2% (\$190,000)
E METHUEN / SWAN VALLEY 500kV	Rebuild line with higher-capacity conductor	\$355,785,000	2% (\$5,282,000)
PATRICK / ROES CORNER 100kV Line	Rebuild line with higher-capacity conductor	\$2,668,000	100% (\$2,668,000)
MUSKOGEE / MONTPELIER ELECTRIC GENERATING STA 500kV	Rebuild line with higher-capacity conductor	\$146,644,000	1% (\$1,111,000)
OMNI-HACKBERRY / NORWICH 500kV	Rebuild line with higher-capacity conductor	\$88,909,000	2% (\$1,416,000)
SWAN VALLEY / TREGO 500kV	Rebuild line with higher-capacity conductor	\$167,267,000	1% (\$1,579,000)
MUSKOGEE / SWAN VALLEY 500kV	Rebuild line with higher-capacity conductor	\$131,257,000	1% (\$958,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SWAN VLY 500/230kV Transformer	Replace transformer with higher-rated unit	\$26,488,000	1% (\$339,000)
IMPERIAL / CHAMBERLIN 230kV L	Rebuild line with higher-capacity conductor	\$18,427,000	1% (\$225,000)
STEVENS RD / CONESVILLE STA 230kV Line	Rebuild line with higher-capacity conductor	\$31,601,000	2% (\$630,000)
15TH AVENUE S SUBSTATION / LUCENT 230kV	Rebuild line with higher-capacity conductor	\$15,022,000	1% (\$173,000)
	Total allocated to project		\$28,219,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.